

Trainings ▾

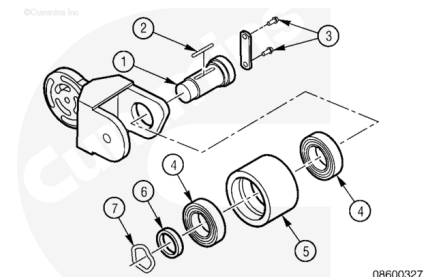
Disassemble

Turnbuckle Idler System

Remove the two capscrews (3) and retaining plate.

Remove the idler shaft (1), idler pulley (5), bearings (4), washer (6), and spring (7).

Remove the key (2) from the idler shaft (1) when present in the idler arm assembly, Part Number 4066540.



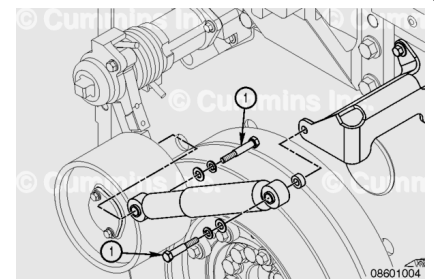
Shock Absorber System

Note : The back side idler system uses one of two types of control rods (turnbuckles) or a shock absorber. Refer to the OEM service manual for the instructions that apply to the engine being serviced.

Remove the back side idler end of the shock absorber, solid control rod (turnbuckle), or control rod tensioner assembly.

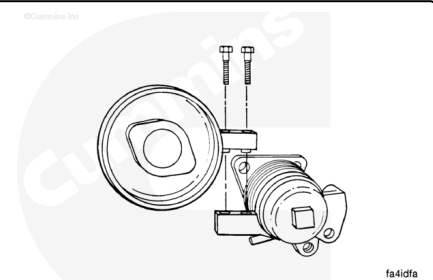
Loosen the upper capscrew (1).

Remove the lower capscrew (1).

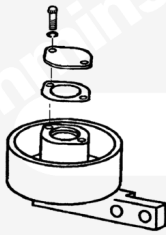


Remove the capscrews.

Remove the pulley assembly from the pivot arm assembly.



Remove the capscrews, lock washers, plate, and gasket.

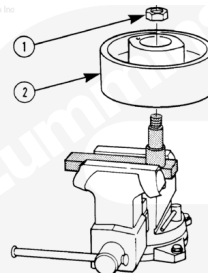


fa6pufa

Note : The locknut has a left-hand thread.

Remove the locknut (1). Lift the pulley (2) from the shaft.

©Cummins Inc.

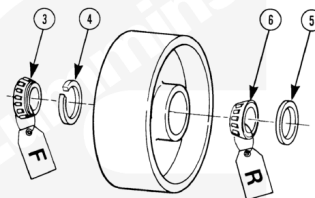


fa4pufc

Remove the inner spacer (3) and rear seal (5).

Remove the front bearing (3) and the rear bearing (6). Tag them for future identification.

©Cummins Inc.



fa4pufd

Clean and Inspect for Reuse

Shock Absorber System

⚠ CAUTION ⚠

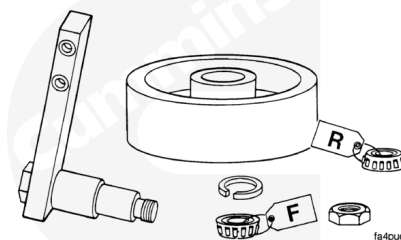
Do not allow the force of the compressed air to turn the bearing when cleaning. Bearing damage will result due to lack of lubrication.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

©Cummins Inc.



fa4puoa

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the parts.

Dry with compressed air.

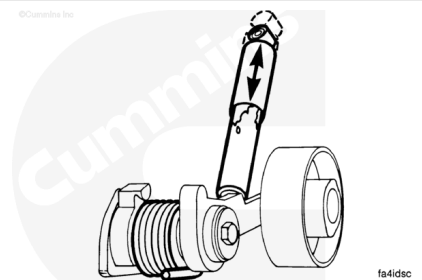
Inspect the front and rear bearings.

The bearings and spacers are part of a kit. See equipment manufacturer service information.

Check the shock absorber for cracks and fluid leakage.

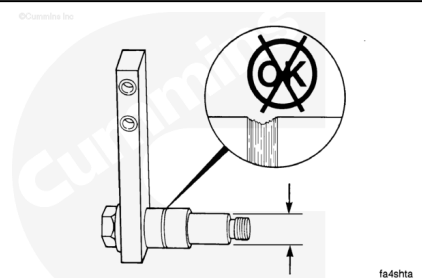
The shock absorber **must** be replaced if the above conditions exist.

Push the shock absorber up and down to check the vibration absorption. Shock absorber should move in and out smoothly.



Inspect the shaft for wear at the seal locations.

Shaft Outside Diameter		
mm		in
21.961	MIN	0.8646
21.974	MAX	0.8651

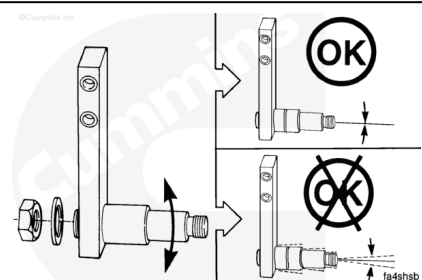


Remove the locknut and lock washer.

Verify the shaft is tight in the bracket. If the shaft is loose, replace both pieces.

Install the parts.

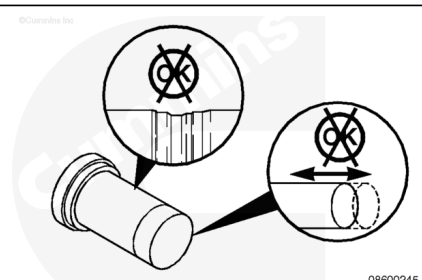
Torque Value: 205 n•m [151 ft-lb]



Turnbuckle Idler System

Check the shaft for wear. Make sure the shaft is tight in the support.

If the shaft has grooves in the surface or is loose, the support and the shaft **must** be replaced as an assembly.



Use idler shaft, Part Number 4001567 or 4066386, and idler bearings, Part Number 4016800, when replacement is necessary.

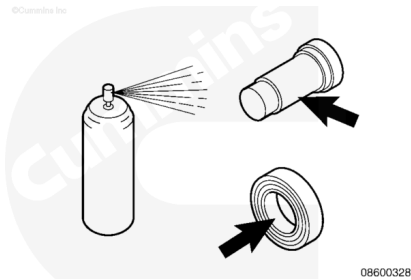
Measure the shaft outside diameter.

Shaft Outside Diameter		
mm		in
59.97	MIN	2.3610
59.98	MAX	2.3614

⚠ CAUTION ⚠

If disassembly is required after the Loctite™ has been applied and the bearings are locked onto the idler shaft, the assembly must be heated to 176.7°C [350°F] to disassemble. The bearings must be replaced after disassembly because this exceeds the approximate rated bearing temperature of 121.1°C [250°F].

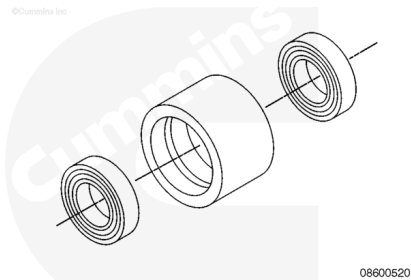
Use Loctite™ primer, Primer T - Loctite™ 7471, Part Number 3824715, to clean the idler shaft and the inside race of the bearings.



Assemble

Turnbuckle Idler System

Install both idler bearings into the idler pulley.



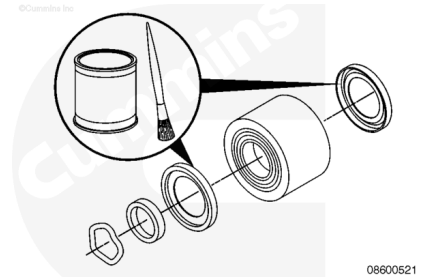
Apply grease in the cavity of the dust shields.

Assemble the dust shields with the greased cavity toward the bearings.

The dust shields **must** align properly with the ball bearing bores.

Install the bearing spacer and spring washer on top of the dust shield.

The components **must** be properly aligned.



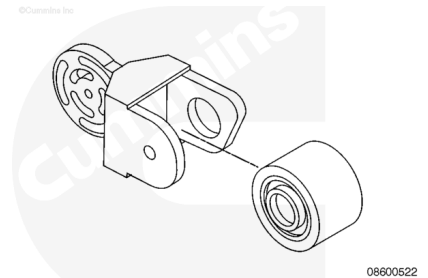
Apply Loctite™ to the inside of the bearing races and idler shaft. Use either of the following bottle sizes of Loctite™ products:

- Loctite™ Threadlocker 242 (Blue) 24221, Part Number 3824041.
- Loctite™ Threadlocker 242 (Blue) 24231, Part Number 3824040.



Install the pulley assembly into the idler lever with the bearing spacer and spring washer facing the smaller bore in the idler lever.

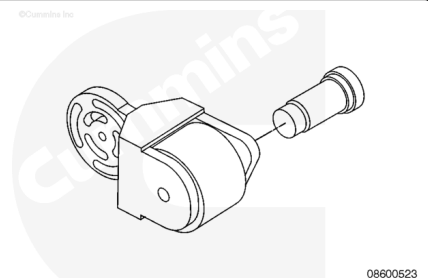
The dust shields, bearing spacer, and spring washer **must** be properly aligned.



Assemble the idler lever, pulley assembly, and idler shaft.

Note : The idler shaft key is **not** needed.

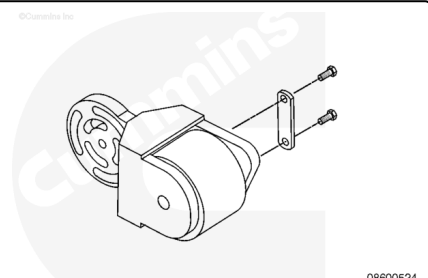
Specific care **must** be taken when inserting the idler shaft so the dust shield next to the bearing spacer remains properly aligned.



Secure the idler shaft into the idler lever and install the retaining plate and capscrews.

Tighten the capscrews.

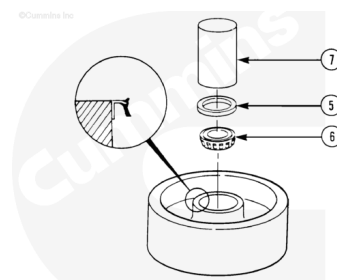
Torque Value: 20 n•m [177 in-lb]



Shock Absorber System



Do not lubricate the seal or the seal surface of the shaft. The seal and seal surface must be clean and dry.



fa4puas

Use high performance, general purpose industrial-type grease (NLGI Grade No. 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or equivalent.

If above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

Lubricate the rear bearing (6) and the bearing race.

Slide the bearing (6) into the bore of the idler pulley.

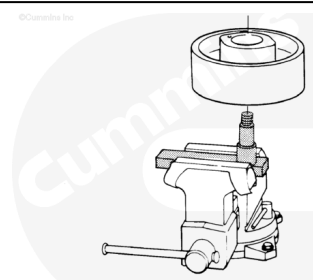
Use a mandrel (7) to install the seal (5).

The seal must be even with the top surface or no more than 0.51 mm [0.020 in] below the top surface.

Install the pulley.

The bearing **must** slide easily over the shaft.

If the bearing does **not** slide easily over the shaft, check for chips and burrs.



fa4puab

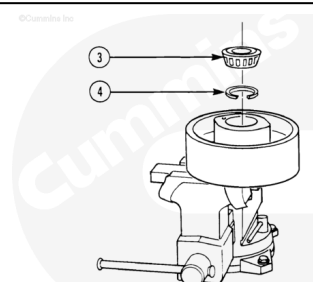
Install the bearing inner spacer (4).

Use high performance, general purpose industrial-type grease (NLGI Grade No. 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or equivalent.

If above greases are **not** available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

Lubricate the front bearing (3) and the bearing race.

Install the bearing (3).

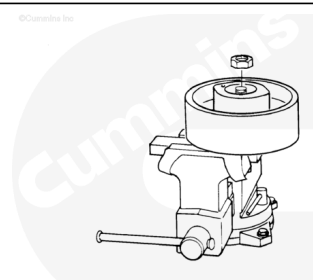


fa4puac

Note : The locknut has a left-hand thread.

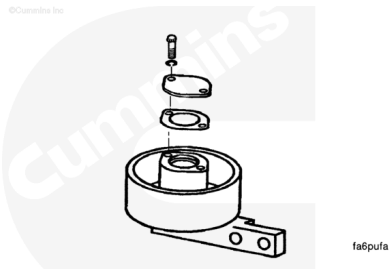
Install and tighten the locknut.

Torque Value: 150 n·m [111 ft-lb]



fa4puad

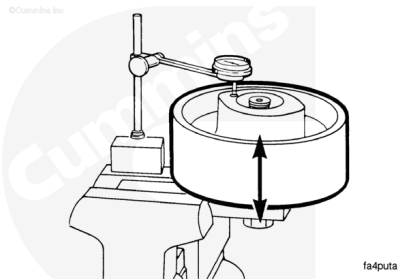
Install the cover plate.



Measure

Use a dial indicator to measure the end clearance.

Bearing End Clearance		
mm		in
0.08	MIN	0.003
0.25	MAX	0.010



If the clearance is **not** within specification, verify the spacer was installed correctly.

The bearings **must** touch the spacer. The bearing races must touch the retaining ring.

Lubricate

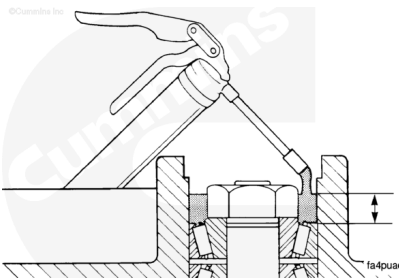
Remove the capscrews, lock washers, cover plate, and gasket.

Use high performance, general purpose industrial-type grease (NLGI Grade No. 2) Chevron™ SRI Grease 2 or Mobilux™ EP 2 grease, or equivalent.

If above greases are not available locally, contact a Cummins® Distributor for advice on other Cummins® approved equivalents.

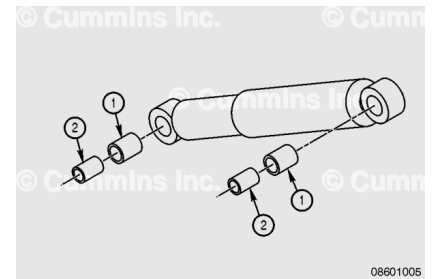
Fill the front cavity 2/3 full of grease.

Install the cover plate.



If the shock absorber has bushing and sleeves supplied separately, then assemble as follows:

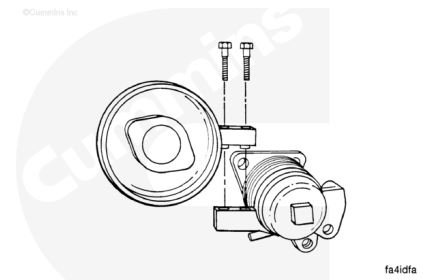
1. Apply P80 Lubriplate™, Cummins® Part Number 3824878, or equivalent, rubber lubricant to the outer surface of the rubber bushing (1). Insert fully into each end of the shock absorber.
2. Apply P80 Lubriplate™, Cummins® Part Number 3822487, or equivalent, rubber lubricant to the outer surface of the sleeve (2). Insert a sleeve into each of the rubber bushings until the sleeve is fully inserted.



Install the pulley assembly onto the pivot arm assembly.

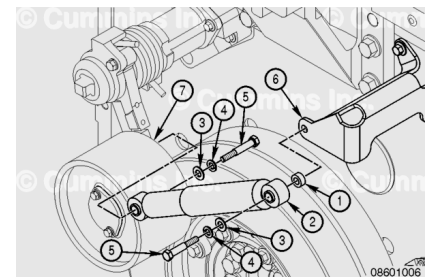
Tighten the capscrews.

Torque Value: 47 n•m [35 ft•lb]



⚠ CAUTION ⚠

The shock absorber must be installed with the larger outer tube of the shock absorber attached to fan hub support. If the shock absorber is installed incorrectly, dirt and debris can enter the tube and cause the component to malfunction.



Install the following part to install the shock absorber:

- Spacer (1)
- Shock absorber (2)
- Flat washer (3)
- Lock washer (4)
- Capscrew (5)

Install the shock absorber (2) in the fan support (6).

Install the flat washer (3), lock washer (4), and capscrew (5) in the lower end of the shock absorber (2).

Install the shock absorber on the fan idler arm (7).

Tighten the two capscrews (5).

